

CORD-19 Research Trends Dashboard

User Guide

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1. Getting Started

The CORD-19 Research Trends Dashboard is a fully interactive executive dashboard that visualizes COVID-19 scientific research trends from 1990 to 2022. No installation or login is required.

How to Access

- Open any web browser (Chrome, Firefox, or Safari)
- Navigate to the following URL:
https://public.tableau.com/app/profile/manoj.kumar.ilango/viz/Book1_projectcapstoneFinal/CORD-19researchTrendsDashboard
- The dashboard loads automatically in your browser
- No Tableau account or software installation required

2. Dashboard Overview

The dashboard is organized into three visual rows:

- Top Row: KPI Summary Cards showing 4 key metrics
- Middle Row: Publication Trend line chart and HeatMap
- Bottom Row: Top Journals, Keywords, Journal Trend, Source Collections, and Yearly Volume charts

Chart	What It Shows	Key Insight
KPI Summary	Total papers, journals, sources, avg/month	499,714 unique papers analyzed
Publication Trend	Monthly research output 1990-2022	COVID spike visible in 2020-2021
HeatMap	Month x year research intensity grid	Dark red = high research activity
Top Journals	Top 20 journals by paper count	BioRxiv leads with ~8,800 papers
Keywords	Top research themes by frequency	sarscov, review, care dominate
Journal Trend	Top 5 journals growth over time	BioRxiv explodes in 2020
Source Collections	Top 10 repositories by papers	Medline/PMC dominates
Yearly Volume	Annual paper count by year	2021 peak: 200,000+ papers

3. How to Use Interactive Features

Pub Year Slider Filter

- Located on the right side of the dashboard
- Drag the LEFT handle to set the start year
- Drag the RIGHT handle to set the end year
- All charts update simultaneously when you adjust the slider
- Example: Drag left handle to 2019 to see only COVID-era research

Journal Filter

- Located on the right side below the Pub Year slider
- Check or uncheck journal names to show/hide them
- Use (All) checkbox to select or deselect all journals at once

Highlight Journal Search

- Type a journal name in the search box to highlight it across all charts
- Example: Type 'Nature' to highlight Nature journal data

Tooltips

- Hover your mouse over any data point to see detailed information
- Tooltips show exact values, journal names, and time periods

4. Common Tasks

How to see only COVID-era research (2019-2023)

- Find the Pub Year slider on the right side
- Drag the left handle from 1990 to 2019
- All charts will update to show only 2019-2023 data

How to find which journal published the most papers

- Look at the Top Journals bar chart in the bottom left
- The longest bar = most papers published
- BioRxiv is the top publisher in this dataset

How to identify the peak research month

- Look at the HeatMap in the middle right
- Find the darkest red square — that is the peak month
- March 2021 had the highest research output

How to explore research themes

- Look at the Keywords treemap in the bottom middle
- Larger boxes = more frequent research themes
- sarscov, review, and care are the top themes

5. Troubleshooting

Dashboard not loading

- Check your internet connection
- Try refreshing the page (Cmd+R on Mac, Ctrl+R on Windows)
- Try a different browser (Chrome, Firefox, Safari)
- Clear browser cache and try again

Charts appear blank

- Wait 10-15 seconds for full load
- Check if Pub Year filter is set to a valid range
- Try resetting the filter by clicking the reset button

For questions contact: Manoj Kumar Ilango | FA 550 Data Visualization | Stevens Institute of Technology | May 2026